# BA ZC581 :
# Applied Data Science
# Module: 03

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- The slides presented here are obtained / extracted from the authors of the prescribed books and from various other subject contributors.
- I hereby thank and acknowledge all the contributors for their material and inputs.
- I have added and modified a few slides to suit the requirements of the course.

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# Module 3: Structured vs Unstructured Data

# Module 3 Topics

Topic 1: Data Types

Topic 2: Internal Data

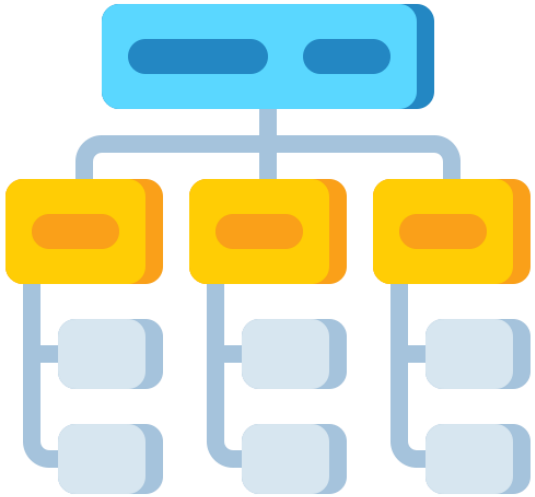
Topic 3: External Data

Topic 4: Semi-Structured Formats

# Data - How it functions in Enterprises

- **Operational Backbone**
  - It powers daily operations—sales transactions, inventory updates, customer interactions, and employee records usually stored in real time databases
- **Strategic Decision Making**
  - Historical and aggregated data supports forecasting, budgeting, and long-term planning aided by warehousing and BI dashboards.
- **Governance & Compliance**
  - Enterprises must manage data securely and ethically, adhering to regulations like GDPR, HIPAA, or India's DPDP Act using data governance frameworks that ensure quality, consistency, and accountability.
- **Integration across Systems**
  - Data flows across ERP, CRM and SCM systems to maintain a unified view of operations. APIs and middleware enable seamless data exchange between platforms.

# A Simple Taxonomy of Data



## By Sourcing

- Internal Data
- External Data

## By Type

- Structured
- Unstructured
- Semi Structured

## By Ownership

- Internally owned
- Externally owned
- Open Data
- Public Data

# Types of Data in Enterprises

- **Structured Data:** Stored in predefined formats like tables—e.g., customer records in a relational database.
- **Semi-Structured Data:** Has some organizational properties—e.g., purchase logs in JSON or XML format.
- **Unstructured Data:** Lacks a fixed format—e.g., emails, scanned invoices, videos, web page contents and social media posts.

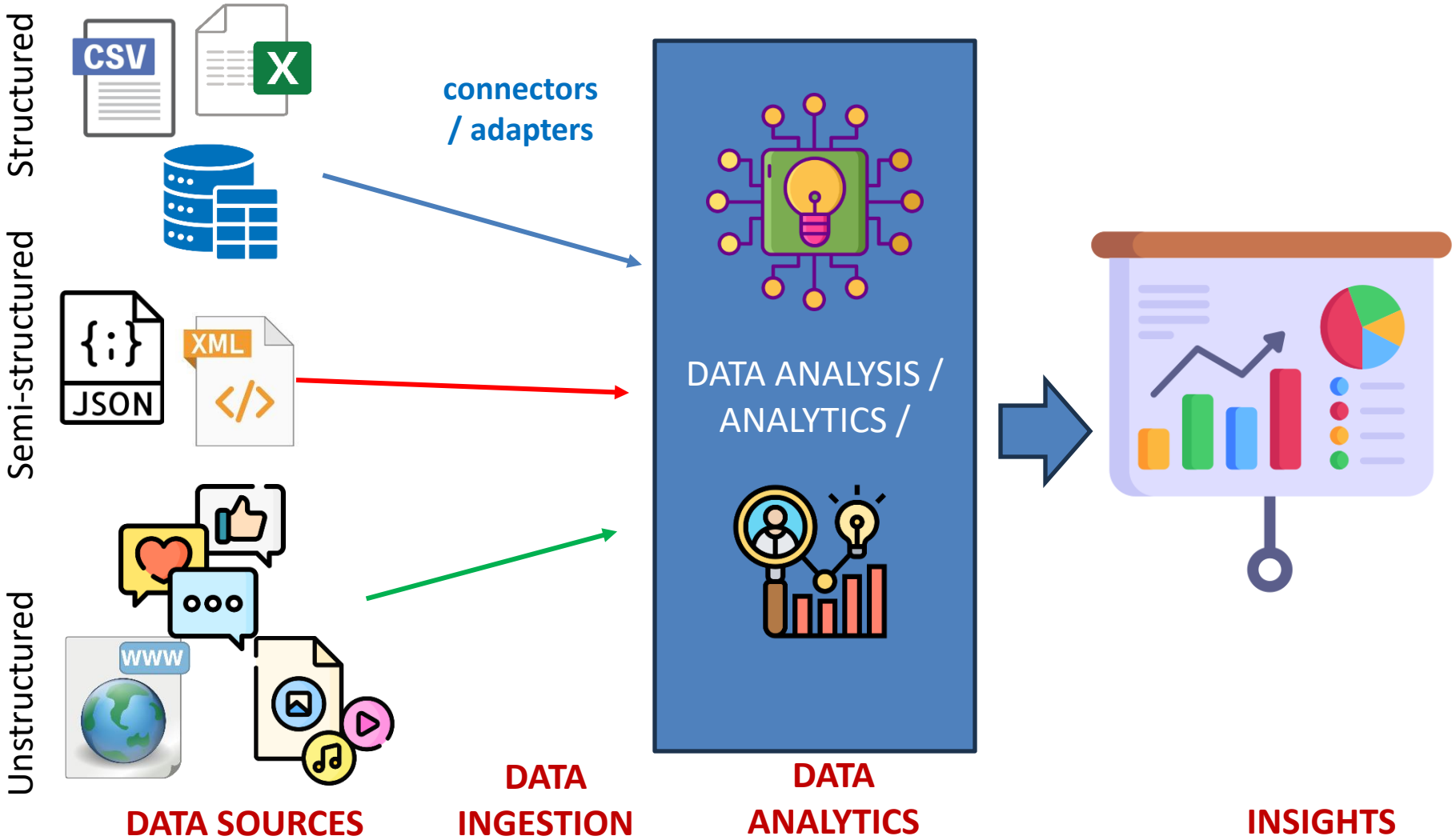
# Unstructured Data

- The biggest Challenge
- Convert to structured / semi structured form
  - ex: convert a text passage into a summarized word; or a sentimental value; a rating number
- Create a quantifiable feature values for the unstructured data
  - ex: a facial image is converted into a feature map

# Some Hard Facts

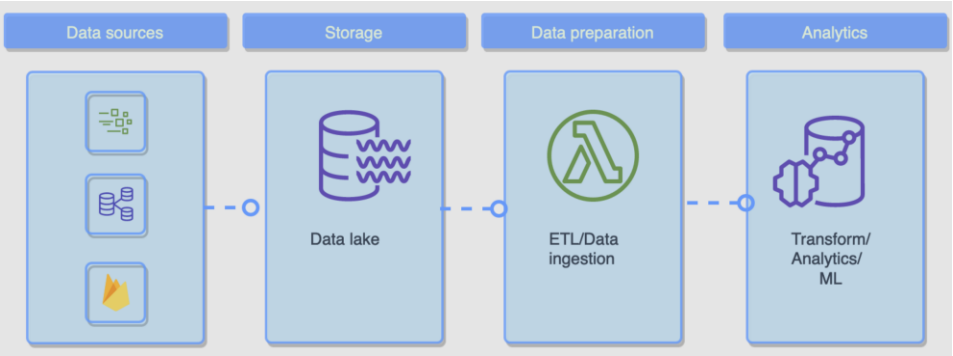
- Gartner states that most of the organization's data and content, between 80% to 90%, is unstructured data, while the rest can be either structured or semi-structured data.
- Major sources are
  - Transactional Data from Databases
  - Document data
  - Sensor (IOT) Data
  - Media data (Images, Video etc)
  - External Data from 3rd party
  - Social Media Data
  - Web data
  - Public Sources

# How do we use data?



# The Data Pipeline

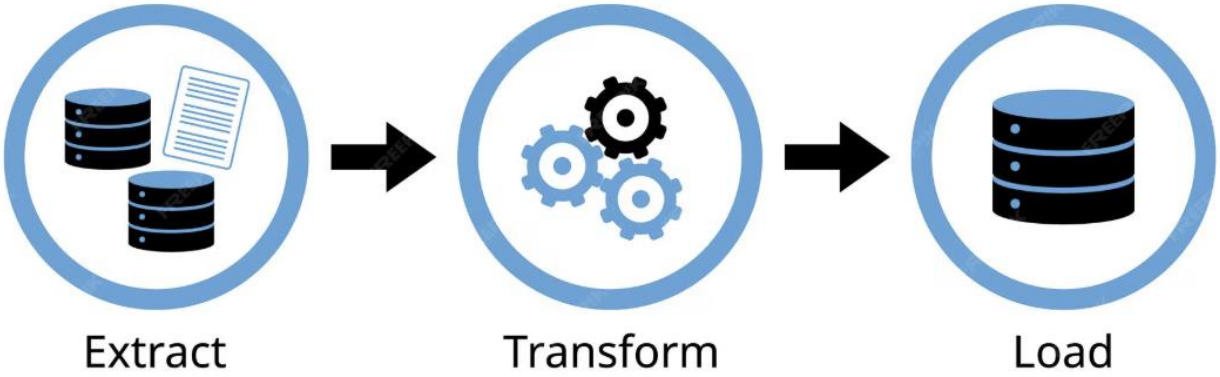
- The end to end process that defines how data flows within enterprise for all data needs.




- Typical functions:
  - Reading from source / previous stage
  - Serialization
  - Transfer
  - Transformations
  - Writing to store / next stage

# The ETL Process

- Important task in the data pipeline
- Various processes carried out to transform data as needed by applications. (mainly to clean and harmonize data)



# Example: Daily Sales Reporting for a Retail Chain

- **Extract :** Data is pulled from multiple sources
  - POS (Point-of-Sale) systems across stores
  - Online order databases
  - Inventory management systems
- **Transform:** Clean and standardize
  - Convert timestamps to a unified format
  - Normalize product codes and categories
  - Calculate total sales, discounts, and taxes
  - Apply business rules:
  - Flag high-value transactions
  - Aggregate sales by region and store
- **Load:** Load the transformed data into a Data Warehouse
  - Used by analysts for daily sales dashboards
  - Supports decision-making on restocking, promotions, and staffing

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# Topic 1: Data Types – A Quick Taxonomy

# A simple picture of Data

- Data: Collection of objects and their attributes representing some facts
- An attribute is a property or characteristic of an object
  - Examples: Designation of an employee, eye color of a person, body temperature, etc.
  - Attribute is also known as variable, field, characteristic, or feature
- A collection of attributes describe an object
  - Object is also known as record, point, sample, entity, or instance

Attributes / fields

Attribute name

| Employee Number | Name | Designation | Base Salary | Reporting To | Experience Years | Qualifications |
|---|---|---|---|---|---|---|
| E305 | Jagdeesh | Director | 45,00,000 | E305 | 25 | Doctorate |
| E308 | Priya | Project Manager | 36,00,000 | E305 | 20 | Masters |
| E310 | Ramesh | Team Leader | 21,00,000 | E308 | 16 | Masters |
| E311 | Swati | Developer | 14,00,000 | E310 | 9 | Graduate |
| E318 | Prasad | Team Leader | 18,00,000 | E308 | 14 | Masters |
| E322 | Surya | Developer | 12,00,000 | E318 | 7 | Graduate |
| E323 | Veena | Developer | 10,00,000 | E318 | 6 | Masters |
| E324 | Mohan | Testing | 9,00,000 | E318 | 6 | Graduate |
| E327 | Kartik | Marketing | 13,00,000 | E308 | 9 | Masters |
| E321 | Rishi | Account | 7,50,000 | E305 | 7 | Masters |

Objects / records

A Record

# Types of Attributes

There are different types of attributes

**Nominal**: Individual objects distinguished from each other

- Examples: ID numbers, eye color, zip codes, Designation

**Ordinal**: Objects which can be ordered

- Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades in exam {A,B,C,D,F}, height as in {tall, medium, short}

**Discrete**: Countable values, often integers

- Examples: number of years of experience, items purchased

**Interval**: Differences between objects is meaningful

- Examples: calendar dates, temperatures in Celsius, Salary.

**Ratio**: Differences and ratios are meaningful

- Examples: temperature in Kelvin, length, time, counts. (temperature in Celsius is not Ratio!)

# Properties of Attribute Values

- The type of an attribute depends on which of the following properties it possesses:

| | |
|---|---|
| Distinctness: | = ≠ |
| Order: | < > |
| Addition: | + - |
| Multiplication: | * / |

- **Nominal attribute**: distinctness
- **Ordinal attribute:** distinctness & order
- **Interval attribute:** distinctness, order & addition
- **Ratio attribute:** all 4 properties

# Qualitative & Quantitative Data

## Qualitative Data

- Descriptive and non-numerical.
- Captures qualities, characteristics, and subjective experiences.
- Examples include interviews, open-ended survey responses, product reviews, and observations.

## Quantitative Data

- Numerical and measurable.
- Involves counts, measurements, and statistical analysis.
- Examples include surveys with closed-ended questions, sales figures, website traffic counts, experimental data, and demographic data.

# Discrete and Continuous Attributes

- Discrete Attribute
  - Has only a finite or countably infinite set of values
  - Examples: zip codes, counts, or the set of words in a collection of documents
  - Often represented as integer variables.
  - Note: binary attributes are a special case of discrete attributes

- Continuous Attribute
  - Has real numbers as attribute values
  - Examples: temperature, height, or weight.
  - Practically, real values can only be measured and represented using a finite number of digits.
  - Continuous attributes are typically represented as floating-point variables.

# Consider these Examples

Consider the following Data representing :

- Military Rank:
  - **Discrete, Qualitative, Ordinal**
- Angles as measured between 0 to 360 degrees:
  - **Continuous, Quantitative, ratio**
- Time in terms of AM or PM.
  - **Binary, Quantitative, Ratio**
- Number of students in any training program
  - **Discrete, Quantitative, Ratio**
- Darkness as measured by Mr. XYZ's point of view.
  - **Discrete, Quantitative, Ordinal**

# Business Data Sources

Main sources of business data include

- **internal** data / database sources like sales records, accounting data, and customer relationship management (CRM) systems, and
- **external** sources such as government reports, social media, and third-party market research.
- Other sources include online analytics, databases, and even manual input from employees or sensors on devices

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# Topic 2: Internal Data

# Internal Data

- Refers to information generated and stored within an organization through its own operations, systems, and processes.
- It is proprietary and typically not accessible to external parties unless shared intentionally

- Feeds into systems like ERP, CRM, HRMS, and BI platforms
- Supports decision-making across departments, enables performance tracking and forecasting, forms the foundation for analytics, reporting, and compliance

# Internal Data

- Available from
  - Transactions entered by operators
  - Data recorded by sensors or machines used in manufacturing process
  - Data stored in internal systems
- Examples:
  - transactional data (customer purchases and staff pay)
  - email marketing metrics (email opens, click rates)
  - information in customer profiles (names, addresses)
  - records of customer interactions (email queries, support calls)
  - online activity (placing items in an online shopping cart)

# Internal Data

- Operational Data
  - **Examples:** Sales transactions, purchase orders, inventory levels, shipping records
  - **Use:** Supports daily business activities and process execution
  - **Systems:** ERP, POS, SCM
- Human Resource Data
  - **Examples:** Employee profiles, attendance logs, payroll records, training history
  - **Use:** Manages workforce planning, compensation, and compliance
  - **Systems:** HRMS, payroll software
- Financial Data
  - **Examples:** Budget reports, expense claims, invoices, tax filings
  - **Use:** Tracks profitability, cash flow, and financial health
  - **Systems:** Accounting software, financial ERP modules
- Strategic & Planning Data
  - **Examples:** Forecasts, market analysis, business plans, performance KPIs
  - **Use:** Guides long-term decisions and resource allocation
  - **Systems:** BI tools, Data warehouse

# Internal Data

- **Sales and transaction data:**
  - Examples: Information from point-of-sale systems, e-commerce platforms, and other sales activities.
  - **Use:** Analyze sales, Predict trends
  - **Systems:** POS, CRM
- Customer Data
  - **Examples:** Understand customer behavior, customer segmentation, CLV analysis
  - **Use:** Segments customer on various aspects, target marketing, customer service effectiveness
  - **Systems:** CRM, Customer service logs, POS
- Operational ERP Data
  - **Examples:** Data from production, procurement, inventory, HR
  - **Use:** Tracks workflow efficiency
  - **Systems:** ERP, SCM, HCM modules
- Market Research Data
  - **Examples:** Data collected firsthand through field surveys, interviews or experiments conducted
  - **Use:** Evaluate market trends, Understand customer requirements, support product development
  - **Systems:** BI tools, CRM

# Internal Data

- Product & Service Data
  - **Examples:** Product specifications, BOMs (Bill of Materials), service logs
  - **Use:** Supports manufacturing, quality control, and customer support
  - **Systems:** PLM, CRM, service management platforms
- Compliance & Governance Data
  - **Examples:** Audit trails, policy documents, regulatory filings
  - **Use:** Ensures legal compliance and internal accountability
  - **Systems:** GRC platforms, document management system

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# Topic 3: External Data

# External Data

- External data refers to information that originates outside an organization and is acquired to support decision-making, strategy, and operations.
- It complements internal data by providing broader context, market intelligence, and competitive insights.

- It includes almost anything from historical demographic data to market prices, or weather conditions to social media trends.
- Organisations use external data to analyze and model economic, political, social, or environmental factors that influence their business

# External Data

- Available from
  - Third party data providers
  - Web and Social media
  - Government authorities

- Examples:
  - Market research reports from agencies like Nielsen or Gartner
  - Public datasets from government portals (e.g., census, trade statistics)
  - Social media trends and customer sentiment from platforms like Twitter or Instagram
  - Competitor pricing and product data scraped from websites or aggregators
  - Industry benchmarks and analyst forecasts
  - News articles and regulatory update

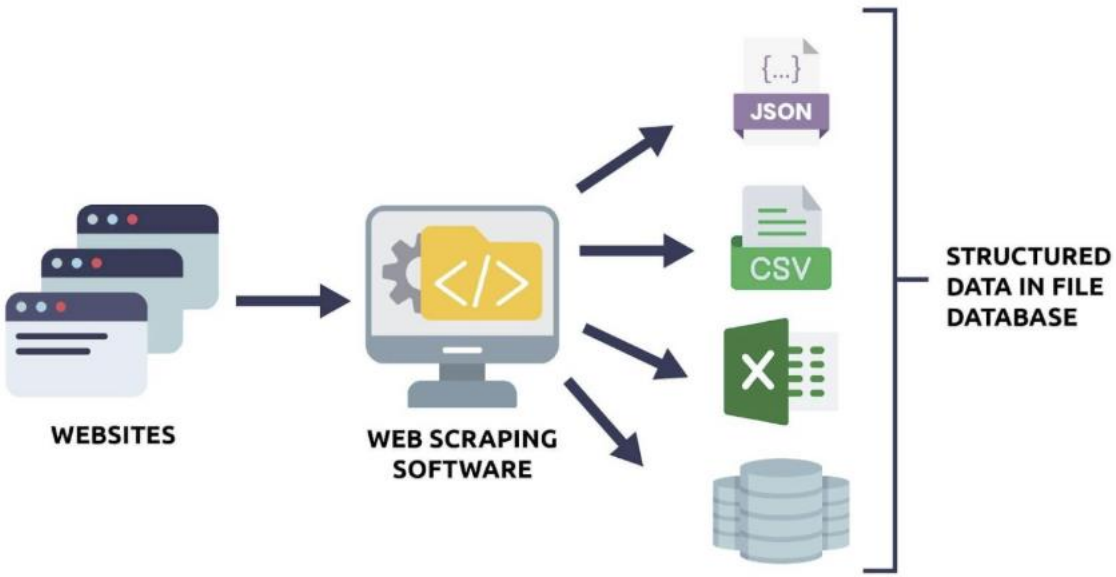
# External Data

- Purpose in Enterprises
  - Enhances strategic planning and forecasting
  - Supports competitive analysis and innovation
  - Allows to gain knowledge of competetion
  - Informs marketing, product development, and risk management
  - Helps comply with external regulations and standard

# Web Scraping for External Data

- Web scraping involves using automated tools or scripts to collect data from websites. This data is then structured and stored for further analysis
- How It Works (ETL-style)
  - **Extract**
    - A scraper (e.g., using Python with BeautifulSoup, Scrapy, or Selenium) visits web pages and pulls HTML content.
  - **Transform**
    - The raw HTML is parsed to extract relevant data like prices, product names, reviews, or news headlines.
    - Data is cleaned, formatted, and structured (e.g., into CSV or JSON).
  - **Load**
    - The cleaned data is stored in databases, data warehouses, or analytics platforms for reporting or modeling

# Web Scraping




Top Web Scraping Tools:

Firecrawl
ScrapeGraphAi
Scrape.do
Browse.Ai
BeautifulSoup
Scrapy
Puppeteer
Playright
Octoparse
ParseHub
Apify

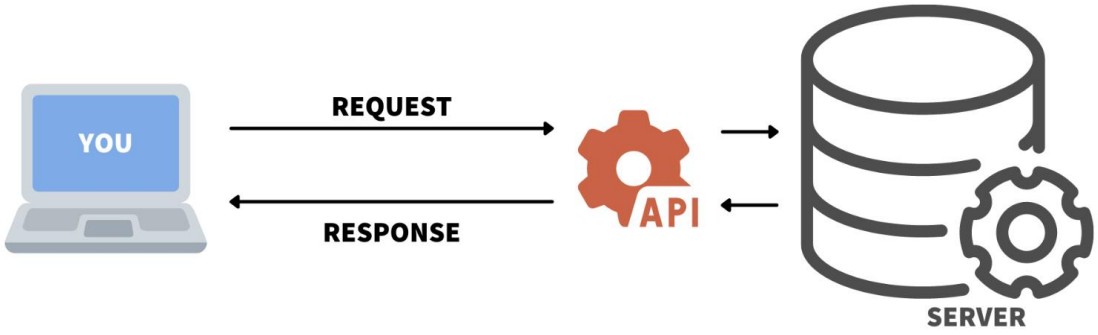
https://www.kdnuggets.com/top-7-ai-web-scraping-tools

# Examples of External Data via Web Scraping

| Use Case | Data Scraped | Purpose |
|---|---|---|
| **Competitive Pricing** | Product prices from competitor websites | Dynamic pricing, promotions |
| **Market Trends** | News articles, blogs, forums | Sentiment analysis, trend spotting |
| **Lead Generation** | Business directories, LinkedIn profiles | Sales prospecting |
| **Real Estate** | Listings from property portals | Market valuation, investment analysis |
| **E-commerce** | Customer reviews, ratings, product specs | Product development, quality tracking |

# APIs for external data

- An **API (Application Programming Interface)** is a set of rules and protocols that allows different software systems to communicate with each other. It acts like a **bridge** that lets one application request data or services from another—often over the internet

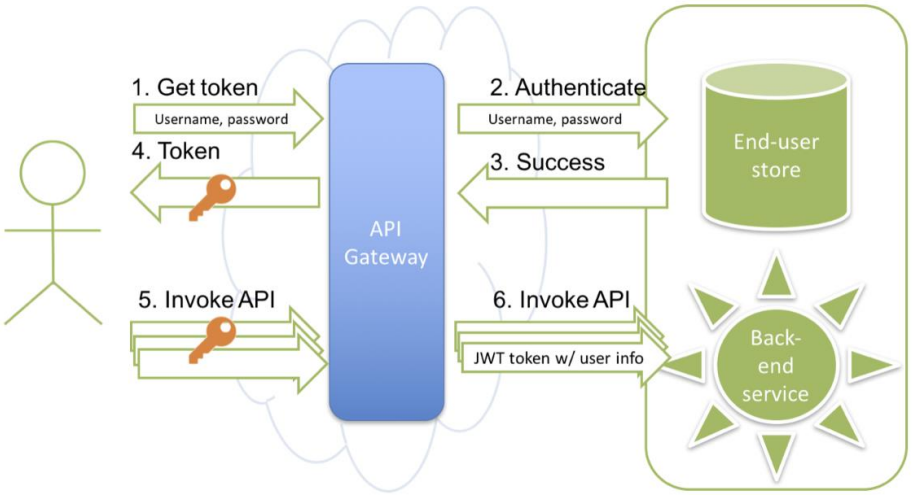


# APIs for external data

- APIs for accessing external data include services for weather, finance, social media, public datasets, and market intelligence—each offering structured endpoints to retrieve real-time or historical information
- A retail app might use a **payment gateway API** to process transactions or a **weather API** to adjust promotions based on local conditions.

# APIs for using external Service

- **Token**: An encrypted value used to authorize and authenticate access to an API to consume a remote service.
- Acts like a **digital key** that tells the API server who you are and what you're allowed to do.




**Bearer Tokens**: Simple and widely used for REST APIs
**OAuth Tokens:** Used for delegated access (e.g., logging in via Google)
**JWT (JSON Web Tokens):** Encodes user info and permissions securely

# Popular APIs

- **Weather and Environmental Data**
  - **OpenWeatherMap API**: Real-time and forecast weather data
  - **Weatherstack API**: Historical and current weather conditions
  - **AirVisual API**: Air quality and pollution metrics
- **Financial and Market Data**
  - **Alpha Vantage API**: Stock prices, forex, and technical indicators
  - **CoinGecko API**: Cryptocurrency prices and market stats
  - **Quandl API**: Economic and financial datasets
- **Social Media and Sentiment**
  - **Twitter API (X API)**: Tweets, trends, user data
  - **Reddit API**: Posts, comments, subreddit analytics
  - **YouTube Data API**: Video metadata, comments, channel stats
- **Public and Government Datasets**
  - **Data.gov API**: U.S. government datasets (health, education, energy)
  - **European Data Portal API**: EU-wide public data
  - **World Bank API**: Global development indicat

# Popular APIs

- **E-commerce and Product Intelligence**
  - **Amazon Product Advertising API**: Product details, reviews, pricing
  - **eBay API**: Listings, categories, seller data
  - **Rakuten RapidAPI Hub**: Aggregated access to multiple e-commerce APIs
- **Market Research and Analytics**
  - **Statista API**: Industry statistics and reports
  - **SimilarWeb API**: Website traffic and engagement metrics
  - **GfK API**: Consumer behavior and retail insights
- **Science, Education, and Research**
  - **NASA API**: Space missions, imagery, and astronomy data
  - **PubMed API**: Biomedical literature and citations
  - **UNESCO Institute for Statistics API**: Global education da

# REST APIs

- REST APIs (Representational State Transfer APIs) are a type of web service that allow systems to communicate over HTTP using simple, standardized operations.

- Stateless: Each request is independent and contains all necessary information
- Uses HTTP methods:
  - GET (retrieve data)
  - POST (create data)
  - PUT (update data)
  - DELETE (remove data)
- Data format: Typically uses JSON or XML
- Resource-based: Accesses data via URLs that represent resources (e.g., /users/123)
- Example: (retrieve product data whose ID=42)

  GET https://api.example.com/products/42

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# Topic 4: Semi Structured Formats

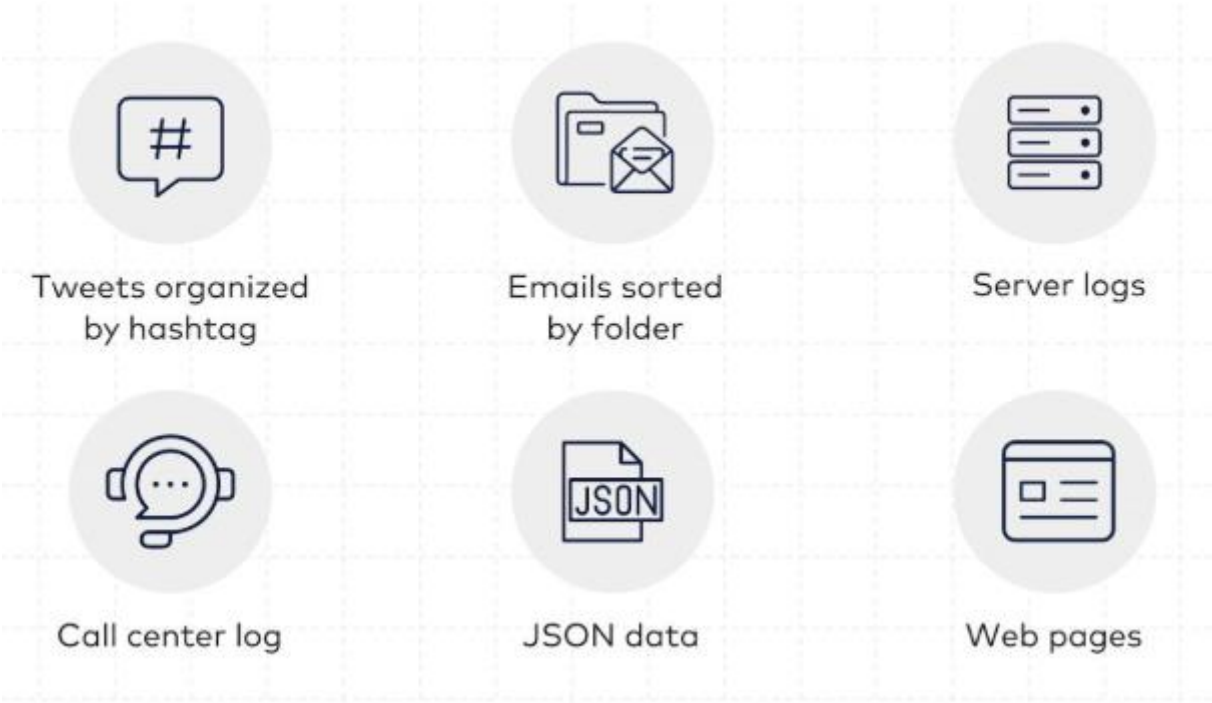
# Semi Structured Formats of Data

- Semi-structured data is a type of data that doesn't fit neatly into traditional relational tables but still contains organizational markers like tags or keys that make it easier to parse than raw unstructured data.

- Characteristics of Semi-structured Data
  - Not stored in fixed rows and columns
  - No formal structure
  - Uses identifiable **tags**, **attributes**, or **hierarchies** to organize data
  - Easier to analyze than unstructured data, but more flexible than structured data

# Semi structured data

# Popular semi structured data



# Common formats of Semi Structured Data

| Format | Description | Example |
|---|---|---|
| **JSON (JavaScript Object Notation)** | Lightweight, human-readable format using key-value pairs | {"name": "Gokhale", "role |
| **XML (eXtensible Markup Language)** | Uses nested tags to define data structure | <user><name>Gokhale< |
| **YAML (YAML Ain't Markup Language)** | Human-friendly format often used in configuration files | name: Gokhale\nrole: De |
| **HTML (HyperText Markup Language)** | Web page structure with embedded data | Gok |
| **NoSQL Documents** | Stored in document databases like MongoDB | JSON-like documents with flexible schemas |
| **Email Metadata** | Headers and tags like To, From, Subject | From: gokhale@example |

# Example json data for an employee


```
{
  "employee_id": "EMP1024",
  "first_name": "Aarav",
  "last_name": "Gokhale",
  "date_of_birth": "1990-06-15",
  "gender": "Male",
  "position": "Curriculum Designer",
  "department": "Education & Innovation",
  "employment_type": "Full-Time",
  "date_of_joining": "2022-01-10",
  "skills": [
    "Curriculum planning",
    "Python",
    "Machine Learning",
    "IoT prototyping",
    "STEAM integration"
  ],
  "contact": {
    "email": "aarav.gokhale@example.com",
    "phone": "+91-9876543210",
    "address": {
      "street": "501 Innovation Lane",
      "city": "Thane",
      "state": "MH",
      "postal_code": "400601",
      "country": "India"
    }
  },
  "status": "Active",
  "performance_rating": 4.8,
  "projects": [
    {
      "project_id": "PRJ2023-STEAM01",
      "name": "STEAM Matrix Deployment",
      "role": "Lead Architect",
      "start_date": "2023-02-01",
      "end_date": "2023-08-15"
    },
    {
      "project_id": "PRJ2024-IOT02",
      "name": "IoT Block Programming Prototype",
      "role": "Systems Integrator",
      "start_date": "2024-01-05",
      "end_date": null
    }
  ]
}
```

# Example Data representation

## Data representation in the 3 types

**Unstructured data**

The university has 5600 students.
John's ID is number 1, he is 18 years old and already holds a B.Sc. degree.
David's ID is number 2, he is 31 years old and holds a Ph.D. degree. Robert's ID is number 3, he is 51 years old and also holds the same degree as David, a Ph.D. degree.

**Semi-structured data**

```
<University>
  <Student ID="1">
    <Name>John</Name>
    <Age>18</Age>
    <Degree>B.Sc.</Degree>
  </Student>
  <Student ID="2">
    <Name>David</Name>
    <Age>31</Age>
    <Degree>Ph.D. </Degree>
  </Student>
....
</University>
```

**Structured data**

| ID | Name | Age | Degree |
|---|---|---|---|
| 1 | John | 18 | B.Sc. |
| 2 | David | 31 | Ph.D. |
| 3 | Robert | 51 | Ph.D. |
| 4 | Rick | 26 | M.Sc. |
| 5 | Michael | 19 | B.Sc. |

# A quick Comparison

| Structured | Semi-structured | Unstructured |
|---|---|---|
| Data is well organized. | Data is organized to some extent. | Data is fully non organized. |
| Organized by the means of relational databases. | Partially organized, e.g. by XML/RDF. | Based on character and binary data. |
| Matured transaction, multiple concurrency techniques. | Transaction is adapted from DBMS, but data concurrency can pose problems. | Difficult, but achievable transaction management and data concurrency. |
| Tuples, rows and tables. | Tuples or graphs are possible. | Versioning usually on whole data or chunks. |
| Schema dependent and less flexible. | Data is more flexible than structured. | The most flexible. |
| Query performance is the highest, structured query can be performed allowing complex joins. | Queries over anonymous nodes are possible. | Schema on-read so query performance is the lowest. |

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# Other types

# Other Data types

1. Time Series
2. Spatial
3. Open Data
4. Big Data

# Time Series Data

- Sequential data where each record is a series of observations taken over time (periodic or a-periodic)
- Usually has a time stamp recorded

Average share prices over the month

| Share | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Reliance | 850 | 880 | 914 | 811 | 780 | 795 | 804 | 845 | 899 | 914 | 999 | 916 |
| Wipro | 454 | 510 | 480 | 491 | 462 | 610 | 560 | 592 | 613 | 688 | 712 | 701 |
| ICICI | 267 | 255 | 283 | 295 | 302 | 319 | 292 | 280 | 273 | 261 | 259 | 246 |

# Spatial Data

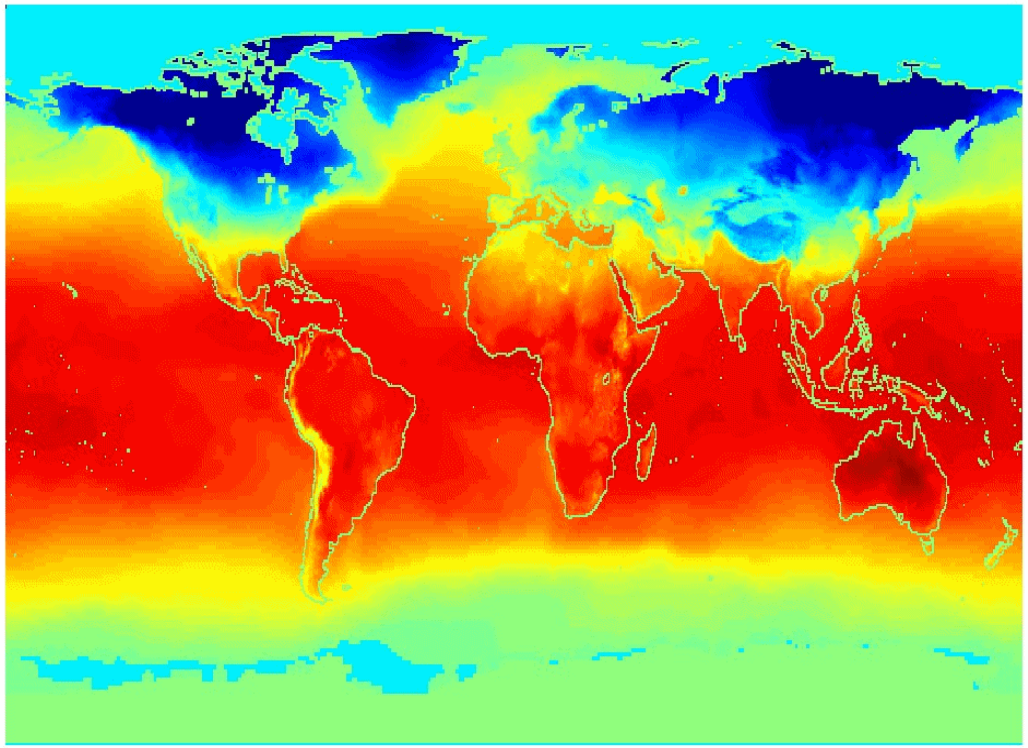
- Data that identifies the location and some characteristics of the location. (also called geo spatial data)
  - Agriculture map, Street map, rainfall map etc

# Spatio-temporal Data

## Depicts data in space and time

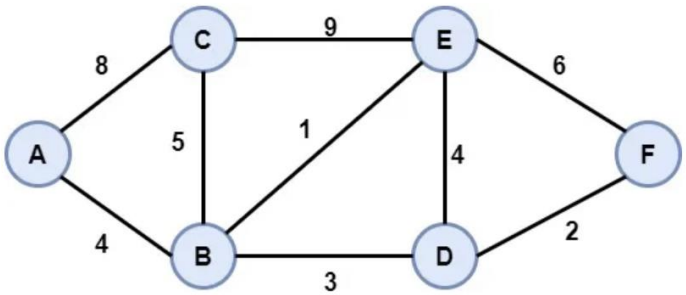
Average Monthly Temperature of land and ocean shown in time and space

Jan

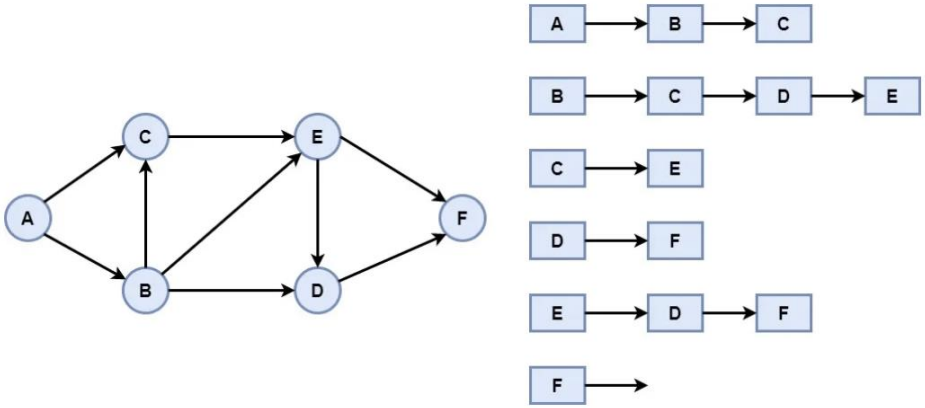


# Graphdata

Identifies inter relationships between data objects (telecom network,



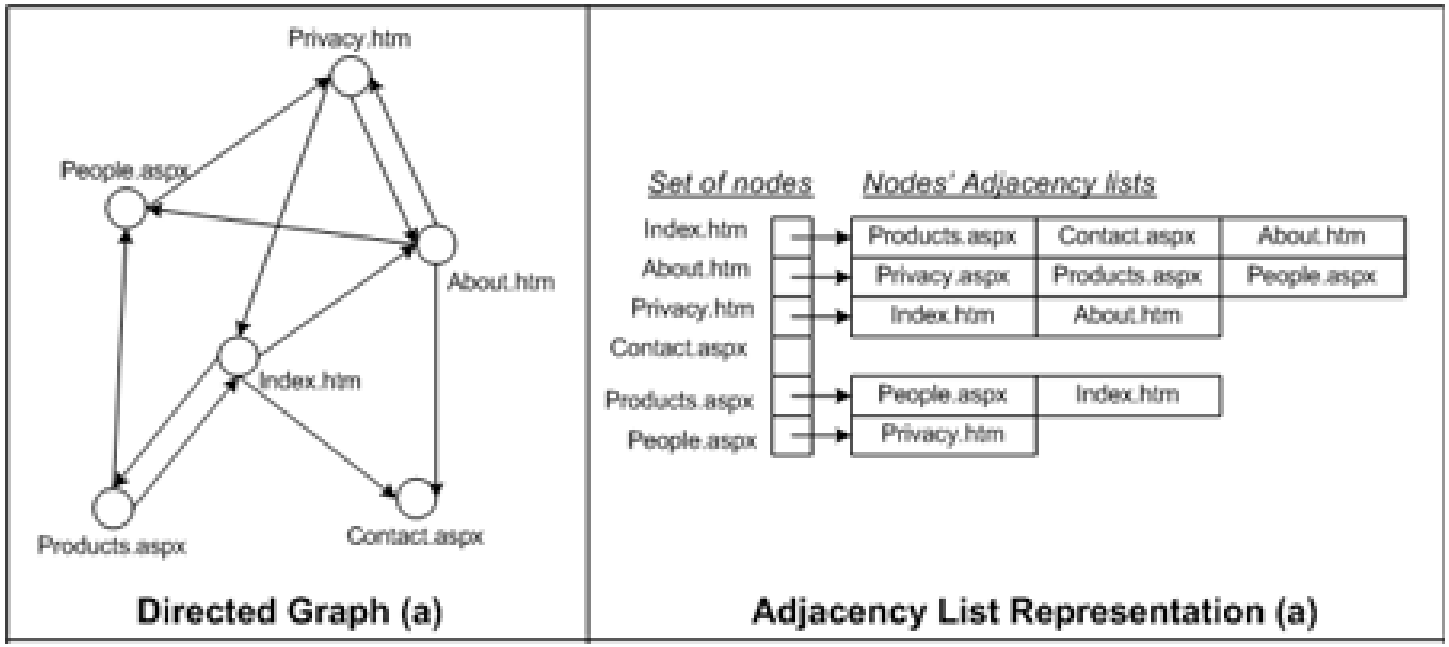

Schematical representation of a graph with weights




Adjacent list of a directed graph

# Example

## Graph data for a web page



# Sequence Data

- Series of values. Sequence data without any timestamp. Ex: gene sequences

```
GGTTCCGCCTTCAGCCCCGCGCC
CGCAGGGCCCGCCCCGCGCCGTC
GAGAAGGGCCCGCCTGGCGGGCG
GGGGGAGGCGGGGCCGCCCGAGC
CCAACCGAGTCCGACCAGGTGCC
CCCTCTGCTCGGCCTAGACCTGA
GCTCATTAGGCGGCAGCGGACAG
GCCAAGTAGAACACGCGAAGCGC
TGGGCTGCCTGCTGCGACCAGGG
```

# Open Data

- Publicly available and free to use
- Often provided by governments or institutions
  - Examples: Census data, public health stats, climate dataset

# Big Data

- Extremely large data (a few petabytes) that needs different techniques to
  - Store (in distributed infrastructure)
  - Process (apply parallel processing techniques)

- The 5Vs of Big Data
  - **Volume**: Big Data involves massive datasets—terabytes to petabytes—generated from sources like sensors, social media, transactions, and logs.
  - **Velocity**: Data is produced and processed at high speed, often in real time (e.g., financial tickers, IoT streams).
  - **Variety**: Includes structured, semi-structured, and unstructured formats—ranging from databases to videos and tweets.
  - **Veracity**: Data quality and reliability can vary, requiring validation and cleansing before use.
  - **Value**: When analyzed effectively, Big Data reveals patterns, trends, and insights that drive strategic decisions.

# Summary

- Discussed various types of data
- Studies about Structured, Unstructured and Semi Structured Data with examples
- Studied the working of APIs

# Questions?